

WT-EXPAND-HW(ADF)环境搭建指南

参考链接：

[espressif/esp-adf: Espressif Audio Development Framework](https://espressif.com/en/esp-adf)

[Get Started - - — Audio Development Framework latest documentation](#)

1、检查适配环境

IDF Version

The following table shows the ESP-IDF versions supported by ESP-ADF at the current time. The label  means supported, and the label  means not supported.

The ESP-IDF master branch is marked as not supported because the major feature changes it has introduced may cause conflicts with ESP-ADF. Yet, the ADF examples not affected by those feature changes can still run correctly on the IDF master branch.

End of Life IDF branches are removed. See [IDF Supported Periods](#) for details.

	ESP-IDF Release/v5.0	ESP-IDF Release/v5.1	ESP-IDF Release/v5.2	ESP-IDF Release/v5.3	ESP-IDF Master
WT-EXPAND-HW Main					
ESP-ADF Master					
ESP-ADF Release/v2.7					
ESP-ADF Release/v2.6					

可以看到，最新的IDF是不支持WT-EXPAND-HW(ADF)的，所以先下载合适的EDP-IDF，本文选择了V5.3.2版本

2、下载安装ESP_IDF v5.3.2

注意：

- 本文在安装V5.3版本时，已经安装了ESP-IDF 其他版本,所以省略了一部分步骤；
- 本文是在虚拟机上操作；
- 本文还在子系统WSL环境以相同步骤实验搭建成功；

如果当前只Ubuntu系统，还没有安装过任何IDF版本，请先执行一下命令，安装ESP-IDF相关应用：

```
sudo apt-get install git wget flex bison gperf python3 python3-pip python3-setuptools cmake ninja-build ccache libffi-dev libssl-dev dfu-util libusb-1.0-0
```

注意：以上命令没有换行，只是因为命令太长文档的自动换行，在执行命令时，需保持完整性；

下载完成后，为了后续使用的方便，创建一下python3的软链接。

```
sudo ln -s /usr/bin/python3 /usr/bin/python
```

创建完成后，可以在命令行中输入“python”来测试是否链接成功。

创建一个名为“ESP_5_3_2”的文件夹

```
mkdir ESP_5_3_2
```

进入“esp”文件夹

```
cd ESP_5_3_2
```

下载 下载工具

```
git clone https://gitee.com/EspressifSystems/esp-gitee-tools.git
```

下载指定版本的esp-idf，在git clone 后面加上 -b （版本号）即可

```
git clone -b v5.3.2 https://gitee.com/EspressifSystems/esp-idf.git
```

在ESP_5_3_2目录，可以看到有两个目录

```
qiming@QIMING20250226:~/ESP_5_3_2$ ls
esp-gitee-tools  esp-idf
qiming@QIMING20250226:~/ESP_5_3_2$ |
```

进入esp-idf

```
cd esp-idf
```

在esp-idf文件夹里使用 下载工具 下载子模块

```
../esp-gitee-tools/submodule-update.sh
```

在esp-idf文件夹里使用工具下载工具链

```
../esp-gitee-tools/install.sh
```

当出现“All done!”字样就代表安装完成。

执行：

```
./export.sh
```

```
Detected installed tools that are not currently used by active ESP-IDF version.
For removing old versions of openocd-esp32, xtensa-esp-elf, esp-rom-elfs, riscv32-esp-elf-gdb, riscv32-esp-elf use co
mmand 'python /home/qiming/5_3_esp/esp-idf/tools/idf_tools.py uninstall'
To free up even more space, remove installation packages of those tools. Use option 'python3 /home/qiming/5_3_esp/esp
-idf/tools/idf_tools.py uninstall --remove-archives'.
```

```
Done! You can now compile ESP-IDF projects.
Go to the project directory and run:
```

```
idf.py build
```

安装完毕，保持窗口；

3、下载安装WT-EXPAND-HW(ADF)

在home目录下下载WT-EXPAND-HW(ADF)

```
git clone https://github.com/FAOEYOI/WT-EXPAND-HW.git
```

```
qiming@QIMING20250226:~/WT-EXPAND-HW$ git clone https://github.com/FAOEYOI/WT-EXPAND-HW.git
Cloning into 'WT-EXPAND-HW'...
remote: Enumerating objects: 18291, done.
remote: Counting objects: 100% (246/246), done.
remote: Compressing objects: 100% (199/199), done.
remote: Total 18291 (delta 101), reused 179 (delta 42), pack-reused 18045 (from 1)
Receiving objects: 100% (18291/18291), 130.61 MiB | 7.01 MiB/s, done.
Resolving deltas: 100% (12556/12556), done.
qiming@QIMING20250226:~/WT-EXPAND-HW$ ls
WT-EXPAND-HW
```

进入WT-EXPAND-HW目录

```
cd WT-EXPAND-HW
```

```
qiming@QIMING20250226:~/WT-EXPAND-HW$ cd WT-EXPAND-HW/
qiming@QIMING20250226:~/WT-EXPAND-HW/WT-EXPAND-HW$ ls
CMakeLists.txt  README.md  conftest.py  esp-idf  export.bat  idf_patches  install.sh  project.mk  tools
LICENSE         components docs        examples  export.sh  install.bat  micropython_adf  pytest.ini
```

进入components目录

```
qiming@QIMING20250226:~/WT-EXPAND-HW/WT-EXPAND-HW/examples/AD35_S3_EXAMPLES/LVGL_DEMO_Music_Player$ ls
CMakeLists.txt  PNG          build        dependencies.lock  managed_components  sdkconfig.defaults  sdkconfig.old
Makefile        README_CN.md components    main              sdkconfig           sdkconfig.defaults.esp32s3
qiming@QIMING20250226:~/WT-EXPAND-HW/WT-EXPAND-HW/examples/AD35_S3_EXAMPLES/LVGL_DEMO_Music_Player$ cd ~/WT-EXPAND-HW/WT-EXPAN
D-HW/components/
qiming@QIMING20250226:~/WT-EXPAND-HW/WT-EXPAND-HW/components$ ls
adf_utils  audio_pipeline  battery_service  display_service  esp_actions  esp_peripherals  tone_partition
audio_board  audio_recorder  bluetooth_service  dueros_service  esp_codec_dev  input_key_service  wifi_service
audio_hal  audio_sal  clouds  esp-adf-libs  esp_dispatcher  ota_service
audio_mixer  audio_stream  coredump_upload_service  esp-sr  esp_event_cast  playlist
qiming@QIMING20250226:~/WT-EXPAND-HW/WT-EXPAND-HW/components$ |
```

删除目录下的那两个文件

```
sudo rm -r esp-sr/
sudo rm -r esp-adf-libs/
```

从github直接克隆下载这两个文件

```
git clone https://github.com/espressif/esp-adf-libs.git
git clone -b v1.9.5 https://github.com/espressif/esp-sr.git
```

最后需要在WT-EXPAND-HW根目录执行

```
./export
```

```
qiming@QIMING20250226:~/WT-EXPAND-HW/WT-EXPAND-HW$ ./export.sh
dirname: invalid option -- 'b'
Try 'dirname --help' for more information.
ADF_PATH: /home/qiming/WT-EXPAND-HW/WT-EXPAND-HW
IDF_PATH: /home/qiming/ESP_5_3_2/esp-idf
Detecting the Python interpreter
Checking "python3" ...
Python 3.10.12
"python3" has been detected
Checking Python compatibility
Checking other ESP-IDF version.
Adding ESP-IDF tools to PATH...
Checking if Python packages are up to date...
Constraint file: /home/qiming/.espressif/esp-idf.constraints.v5.3.txt
```

.....

```
<string>:1: DeprecationWarning: The '__version__' attribute is deprecated and will be removed in Click 9.1. Use feature detect
ion or 'importlib.metadata.version("click")' instead.
/home/qiming/ESP_5_3_2/esp-idf/tools/idf.py:357: DeprecationWarning: 'MultiCommand' is deprecated and will be removed in Click
9.0. Use 'Group' instead.
  class CLI(click.MultiCommand):
Checking "python3" ...
Python 3.10.12
"python3" has been detected
Applying patch /home/qiming/WT-EXPAND-HW/WT-EXPAND-HW/idf_patches/esp32p4_hosted_and_sdcard_coexistent.patch
Applying patch /home/qiming/WT-EXPAND-HW/WT-EXPAND-HW/idf_patches/idf_v5.3_freertos.patch

The following command can be executed now to view detailed usage:

idf.py --help

Compilation example (The commands highlighted in yellow below are optional: Configure the chip and project settings separately
)

cd /home/qiming/WT-EXPAND-HW/WT-EXPAND-HW/examples/cli
idf.py set-target esp32
idf.py menuconfig
idf.py build
```

看到如图所示，说明环境安装并成功运行；

到这里就基本没有问题了

随便找个 WT-EXPAND-HW 的examples目录里的例子编译烧录测试一个试试

4、测试WT-EXPAND-HW(ADF)环境

进入项目，设置目标芯片(确保已执行前面的步骤)

图片所示例程路径：WT-EXPAND-HW/examples/AD35_S3_EXAMPLES/LVGL_DEMO_Music_Player

```
qiming@QIMING20250226:~/WT-EXPAND-HW/WT-EXPAND-HW$ ls
CMakeLists.txt  README.md  conftest.py  esp-idf      export.bat  idf_patches  install.sh    project.mk  tools
LICENSE         components docs         examples     export.sh   install.bat  micropython_adf  pytest.ini
qiming@QIMING20250226:~/WT-EXPAND-HW/WT-EXPAND-HW$ cd examples/
qiming@QIMING20250226:~/WT-EXPAND-HW/WT-EXPAND-HW/examples$ ls
AD35_S3_EXAMPLES  advanced_examples  cli          dueros      ota          recorder
README.md         audio_processing   cloud_services  get-started  player       speech_recognition
README_CN.md      checks            display       korvo-du1906  protocols    system
qiming@QIMING20250226:~/WT-EXPAND-HW/WT-EXPAND-HW/examples$ cd AD35_S3_EXAMPLES/
qiming@QIMING20250226:~/WT-EXPAND-HW/WT-EXPAND-HW/examples/AD35_S3_EXAMPLES$ ls
Aw9523B_GET_ID  BACK_IIC_AHT20_TEST  BACK_UART_TEST  MK2016_Proximity_Sensor_TEST  TOUCH_FT5X06_TEST
BACK_IIC_AHT20_TEST  LCD_ST7796_TEST    Play_Music_TEST  mcpwm_servo_control
BACK_SPI_TEST      LVGL_DEMO_Music_Player  Recording_playback_TEST
qiming@QIMING20250226:~/WT-EXPAND-HW/WT-EXPAND-HW/examples/AD35_S3_EXAMPLES$ cd LVGL_DEMO_Music_Player/
qiming@QIMING20250226:~/WT-EXPAND-HW/WT-EXPAND-HW/examples/AD35_S3_EXAMPLES/LVGL_DEMO_Music_Player$ ls
CMakeLists.txt  PNG          build      dependencies.lock  managed_components  sdkconfig.defaults  sdkconfig.old
Makefile        README_CN.md  components  main               sdkconfig            sdkconfig.defaults.esp32s3
```

执行以下命令进入配置界面，设置目标开发板：

```
idf.py menuconfig
```

```
(Top) → Audio HAL → Audio board
Espressif IoT Development Framework Configuration
( ) Custom audio board
(X) Audio board AD35-S3
( ) ESP32-Lyrat V4.3
( ) ESP32-Lyrat V4.2
( ) ESP32-LyraTD-MSC V2.1
( ) ESP32-LyraTD-MSC V2.2
( ) ESP32-Lyrat-Mini V1.1
( ) ESP32_KORVO_DU1906
( ) ESP32-S2-Kaluga-1 v1.2
( ) ESP32-S3-Korvo-2 v3
( ) ESP32-S3-Korvo-2L v1
( ) ESP32-S3-BOX-Lite
( ) ESP32-S3-BOX
( ) ESP32-S3-BOX-3
( ) ESP32-C3-Lyra-v2.0
( ) ESP32-C6-DEVKIT
( ) ESP32-P4-FUNCTION-EV-BOARD

[Space/Enter] Toggle/enter  [ESC] Leave menu      [S] Save
[O] Load                  [?] Symbol info      [/] Jump to symbol
[F] Toggle show-help mode  [C] Toggle show-name mode  [A] Toggle show-all mode
[Q] Quit (prompts for save) [D] Save minimal config (advanced)
```

【S】保存，【Q】退出；

编译：

```

qiming@QIMING20250226:~/WT-EXPAND-HW/WT-EXPAND-HW/examples/AD35_S3_EXAMPLES/LVGL_DEMO_Music_Player$ ls
CMakeLists.txt  PNG          build        dependencies.lock  sdkconfig        sdkconfig.defaults.esp32s3
Makefile        README_CN.md  components  main              sdkconfig.defaults  sdkconfig.old
qiming@QIMING20250226:~/WT-EXPAND-HW/WT-EXPAND-HW/examples/AD35_S3_EXAMPLES/LVGL_DEMO_Music_Player$ idf.py build
/home/qiming/ESP_5_3_2/esp-idf/tools/idf.py:357: DeprecationWarning: 'MultiCommand' is deprecated and will be removed in Click
9.0. Use 'Group' instead.
  class CLI(click.MultiCommand):
Executing action: all (aliases: build)
Running cmake in directory /home/qiming/WT-EXPAND-HW/WT-EXPAND-HW/examples/AD35_S3_EXAMPLES/LVGL_DEMO_Music_Player/build
Executing "cmake -G Ninja -DPYTHON_DEPS_CHECKED=1 -DPYTHON=/home/qiming/.espressif/python_env/idf5.3_py3.10_env/bin/python -DE
SP_PLATFORM=1 -DCCACHE_ENABLE=0 /home/qiming/WT-EXPAND-HW/WT-EXPAND-HW/examples/AD35_S3_EXAMPLES/LVGL_DEMO_Music_Player"...
-- IDF_TARGET is not set, guessed 'esp32s3' from sdkconfig '/home/qiming/WT-EXPAND-HW/WT-EXPAND-HW/examples/AD35_S3_EXAMPLES/L
VGL_DEMO_Music_Player/sdkconfig'
-- Found Git: /usr/bin/git (found version "2.34.1")
-- The C compiler identification is GNU 13.2.0
-- The CXX compiler identification is GNU 13.2.0
-- The ASM compiler identification is GNU
-- Found assembler: /home/qiming/.espressif/tools/xtensa-esp-elf/esp-13.2.0_20240530/xtensa-esp-elf/bin/xtensa-esp32s3-elf-gcc
-- Detecting C compiler ABI info
-- Detecting C compiler ABI info - done
-- Check for working C compiler: /home/qiming/.espressif/tools/xtensa-esp-elf/esp-13.2.0_20240530/xtensa-esp-elf/bin/xtensa-es
p32s3-elf-gcc - skipped

```

.....

```

    252 | esp_rom_spiflash_result_t esp_rom_spiflash_read(uint32_t src_addr, uint32_t *dest, int32_t len);
    |
[1557/1558] Generating binary image from built executable
esptool.py v4.8.1
Creating esp32s3 image...
Merged 2 ELF sections
Successfully created esp32s3 image.
Generated /home/qiming/WT-EXPAND-HW/WT-EXPAND-HW/examples/AD35_S3_EXAMPLES/LVGL_DEMO_Music_Player/build/LVGL_DEMO_Music_Player
.bin
[1558/1558] cd /home/qiming/WT-EXPAND-HW/WT-EXPAND-HW/examples/A...EXAMPLES/LVGL_DEMO_Music_Player/build/LVGL_DEMO_Music_Playe
r.bi
LVGL_DEMO_Music_Player.bin binary size 0xd1570 bytes. Smallest app partition is 0x100000 bytes. 0x2ea90 bytes (18%) free.

Project build complete. To flash, run:
idf.py flash
or
idf.py -p PORT flash
or
python -m esptool --chip esp32s3 -b 460800 --before default_reset --after hard_reset write_flash --flash_mode dio --flash_siz
e 16MB --flash_freq 80m 0x0 build/bootloader/bootloader.bin 0x8000 build/partition_table/partition-table.bin 0x10000 build/LVG
L_DEMO_Music_Player.bin
or from the "/home/qiming/WT-EXPAND-HW/WT-EXPAND-HW/examples/AD35_S3_EXAMPLES/LVGL_DEMO_Music_Player/build" directory
python -m esptool --chip esp32s3 -b 460800 --before default_reset --after hard_reset write_flash "@flash_args"

```

烧录：

这里的端口可以通过拔插USB,通过

```
ls /dev/tty*
```

查看设备对应的端口号,假设确认的端口号为: /dev/ttyUSB0

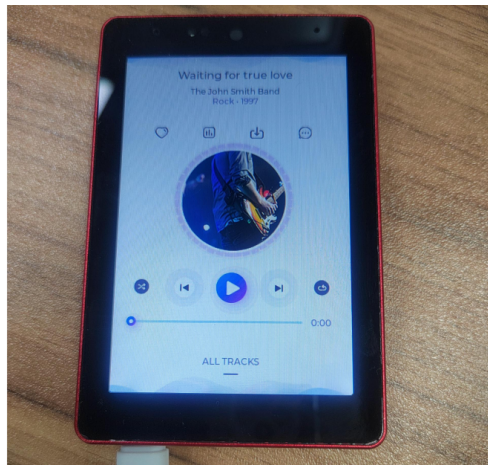
执行以下命令修改权限;

```
sudo chmod 777 /dev/ttyUSB0
```

然后执行:

```
idf.py -p /dev/ttyUSB0 flash monitor
```

测试结果：



在Ubuntu虚拟机，WT-EXPAND-HW(ADF)环境搭建成功!

5、如果是在WSL子系统环境下搭建

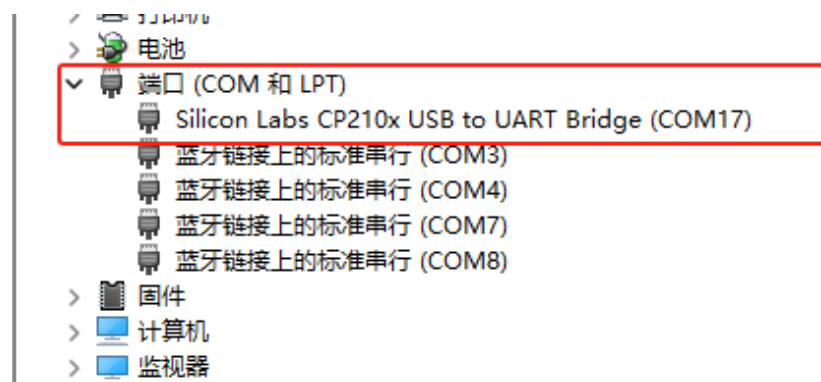
运行WT-EXPAND-HW/examples/get-started/play_mp3_control

5.1 找到端口号

注意:

windows系统中串口设备名称是: COM17, 对应windows子系统中设备名称是: /dev/ttyS17

(如果是COM6, 对应windows子系统中设备名称是: /dev/ttyS6)



```
wan@WANYUJIAN24:~/ESP_5_3/esp-adf/examples/get-started/play_mp3_control$ ls /dev/tty*
/dev/tty      /dev/ttyS115 /dev/ttyS135 /dev/ttyS155 /dev/ttyS175 /dev/ttyS22  /dev/ttyS42  /dev/ttyS62  /dev/ttyS82
/dev/tty0     /dev/ttyS116 /dev/ttyS136 /dev/ttyS156 /dev/ttyS176 /dev/ttyS23  /dev/ttyS43  /dev/ttyS63  /dev/ttyS83
/dev/tty1     /dev/ttyS117 /dev/ttyS137 /dev/ttyS157 /dev/ttyS177 /dev/ttyS24  /dev/ttyS44  /dev/ttyS64  /dev/ttyS84
/dev/ttyS0    /dev/ttyS118 /dev/ttyS138 /dev/ttyS158 /dev/ttyS178 /dev/ttyS25  /dev/ttyS45  /dev/ttyS65  /dev/ttyS85
/dev/ttyS1    /dev/ttyS119 /dev/ttyS139 /dev/ttyS159 /dev/ttyS179 /dev/ttyS26  /dev/ttyS46  /dev/ttyS66  /dev/ttyS86
/dev/ttyS10   /dev/ttyS12  /dev/ttyS14  /dev/ttyS16  /dev/ttyS18  /dev/ttyS27  /dev/ttyS47  /dev/ttyS67  /dev/ttyS87
/dev/ttyS100  /dev/ttyS120 /dev/ttyS140 /dev/ttyS160 /dev/ttyS180 /dev/ttyS28  /dev/ttyS48  /dev/ttyS68  /dev/ttyS88
/dev/ttyS101  /dev/ttyS121 /dev/ttyS141 /dev/ttyS161 /dev/ttyS181 /dev/ttyS29  /dev/ttyS49  /dev/ttyS69  /dev/ttyS89
/dev/ttyS102  /dev/ttyS122 /dev/ttyS142 /dev/ttyS162 /dev/ttyS182 /dev/ttyS3   /dev/ttyS5   /dev/ttyS7   /dev/ttyS9
/dev/ttyS103  /dev/ttyS123 /dev/ttyS143 /dev/ttyS163 /dev/ttyS183 /dev/ttyS30  /dev/ttyS50  /dev/ttyS70  /dev/ttyS90
/dev/ttyS104  /dev/ttyS124 /dev/ttyS144 /dev/ttyS164 /dev/ttyS184 /dev/ttyS31  /dev/ttyS51  /dev/ttyS71  /dev/ttyS91
/dev/ttyS105  /dev/ttyS125 /dev/ttyS145 /dev/ttyS165 /dev/ttyS185 /dev/ttyS32  /dev/ttyS52  /dev/ttyS72  /dev/ttyS92
/dev/ttyS106  /dev/ttyS126 /dev/ttyS146 /dev/ttyS166 /dev/ttyS186 /dev/ttyS33  /dev/ttyS53  /dev/ttyS73  /dev/ttyS93
/dev/ttyS107  /dev/ttyS127 /dev/ttyS147 /dev/ttyS167 /dev/ttyS187 /dev/ttyS34  /dev/ttyS54  /dev/ttyS74  /dev/ttyS94
/dev/ttyS108  /dev/ttyS128 /dev/ttyS148 /dev/ttyS168 /dev/ttyS188 /dev/ttyS35  /dev/ttyS55  /dev/ttyS75  /dev/ttyS95
/dev/ttyS109  /dev/ttyS129 /dev/ttyS149 /dev/ttyS169 /dev/ttyS189 /dev/ttyS36  /dev/ttyS56  /dev/ttyS76  /dev/ttyS96
/dev/ttyS11   /dev/ttyS13  /dev/ttyS15  /dev/ttyS17  /dev/ttyS19  /dev/ttyS37  /dev/ttyS57  /dev/ttyS77  /dev/ttyS97
/dev/ttyS110  /dev/ttyS130 /dev/ttyS150 /dev/ttyS170 /dev/ttyS190 /dev/ttyS38  /dev/ttyS58  /dev/ttyS78  /dev/ttyS98
/dev/ttyS111  /dev/ttyS131 /dev/ttyS151 /dev/ttyS171 /dev/ttyS191 /dev/ttyS39  /dev/ttyS59  /dev/ttyS79  /dev/ttyS99
/dev/ttyS112  /dev/ttyS132 /dev/ttyS152 /dev/ttyS172 /dev/ttyS2   /dev/ttyS4   /dev/ttyS6   /dev/ttyS8
/dev/ttyS113  /dev/ttyS133 /dev/ttyS153 /dev/ttyS173 /dev/ttyS20  /dev/ttyS40  /dev/ttyS60  /dev/ttyS80
/dev/ttyS114  /dev/ttyS134 /dev/ttyS154 /dev/ttyS174 /dev/ttyS21  /dev/ttyS41  /dev/ttyS61  /dev/ttyS81
```

5.2 烧录

```
wan@WANYUJIAN24:~/ESP_5_3/esp-adf/examples/get-started/play_mp3_control$ idf.py -p /dev/ttyS17 flash -b 115200 monitor
```

5.2.1 注意:

在子系统在执行烧录命令时，需要加上波特率，否则会烧录失败：

```
termios.error: (5, 'Input/output error')
FAILED: CMakeFiles/Flash /home/wan/ESP_5_3/esp-adf/examples/get-started/play_mp3_control/build/CMakeFiles/Flash
cd /home/wan/ESP_5_3/esp-idf/components/esptool_py && /usr/bin/cmake -D IDF_PATH=/home/wan/ESP_5_3/esp-idf -D "SERIAL_TOOL=/home/wan/.espressif/python_env/idf5.3_py3.10_env/bin/python;;/home/wan/ESP_5_3/esp-idf/components/esptool_py/esptool/esptool.py;--chip;esp32s3" -D "SERIAL_TOOL_ARGS=--before=default_reset;--after=hard_reset;write_flash;@flash_args" -D WORKING_DIRECTORY=/home/wan/ESP_5_3/esp-adf/examples/get-started/play_mp3_control/build -P /home/wan/ESP_5_3/esp-idf/components/esptool_py/run_serial_tool.cmake
ninja: build stopped: subcommand failed.
CMake Error at run_serial_tool.cmake:66 (message):

  /home/wan/.espressif/python_env/idf5.3_py3.10_env/bin/python;;/home/wan/ESP_5_3/esp-idf/components/esptool_py/esptool/esptool.py;--chip;esp32s3
  failed.
```

5.2.2 成功结果

```
W (398) ADC: legacy driver is deprecated, please migrate to 'esp_adc/adc_oneshot.h'
I (398) sleep: Configure to isolate all GPIO pins in sleep state
I (405) sleep: Enable automatic switching of GPIO sleep configuration
I (412) main_task: Started on CPU0
I (432) main_task: Calling app_main()
I (432) PLAY_FLASH_MP3_CONTROL: [ 1 ] Start audio codec chip
W (432) i2c_bus_v2: I2C master handle is NULL, will create new one
W (482) ES7210: Enable TDM mode. ES7210_SDP_INTERFACE2_REG12: 2
I (492) PLAY_FLASH_MP3_CONTROL: [ 2 ] Create audio pipeline, add all elements to pipeline, and subscribe pipeline event
I (492) PLAY_FLASH_MP3_CONTROL: [2.1] Create mp3 decoder to decode mp3 file and set custom read callback
I (502) PLAY_FLASH_MP3_CONTROL: [2.2] Create i2s stream to write data to codec chip
I (512) PLAY_FLASH_MP3_CONTROL: [2.3] Register all elements to audio pipeline
I (522) PLAY_FLASH_MP3_CONTROL: [2.4] Link it together [mp3_music_read_cb]-->mp3_decoder-->i2s_stream-->[codec_chip]
I (532) PLAY_FLASH_MP3_CONTROL: [ 3 ] Initialize peripherals
I (542) PLAY_FLASH_MP3_CONTROL: [3.1] Initialize keys on board
I (542) PLAY_FLASH_MP3_CONTROL: [ 4 ] Set up event listener
I (552) PLAY_FLASH_MP3_CONTROL: [4.1] Listening event from all elements of pipeline
I (562) PLAY_FLASH_MP3_CONTROL: [4.2] Listening event from peripherals
W (562) PLAY_FLASH_MP3_CONTROL: [ 5 ] Tap touch buttons to control music player:
W (572) PLAY_FLASH_MP3_CONTROL: [Play] to start, pause and resume, [Set] to stop.
W (582) PLAY_FLASH_MP3_CONTROL: [Vol-] or [Vol+] to adjust volume.
I (592) PLAY_FLASH_MP3_CONTROL: [ 5.1 ] Start audio_pipeline
W (602) AUDIO_THREAD: Make sure selected the 'CONFIG_SPIRAM_BOOT_INIT' and 'CONFIG_SPIRAM_ALLOW_STACK_EXTERNAL_MEMORY' by 'make menuconfig'
I (622) PLAY_FLASH_MP3_CONTROL: [ * ] Receive music info from mp3 decoder, sample_rates=8000, bits=16, ch=2
```

在WSL搭建ESP-ADF环境也成功！

5.3 备用烧录方法

部分电脑可能出现找不到端口号的问题，以下是解决方法：

5.3.1 idfx工具的下载

执行命令：

```
git clone https://github.com/abobija/idx.git
```

下载idx工具；

下载完成后将文件名为【idx】的文件放在前面下载的idf环境的tools目录下：

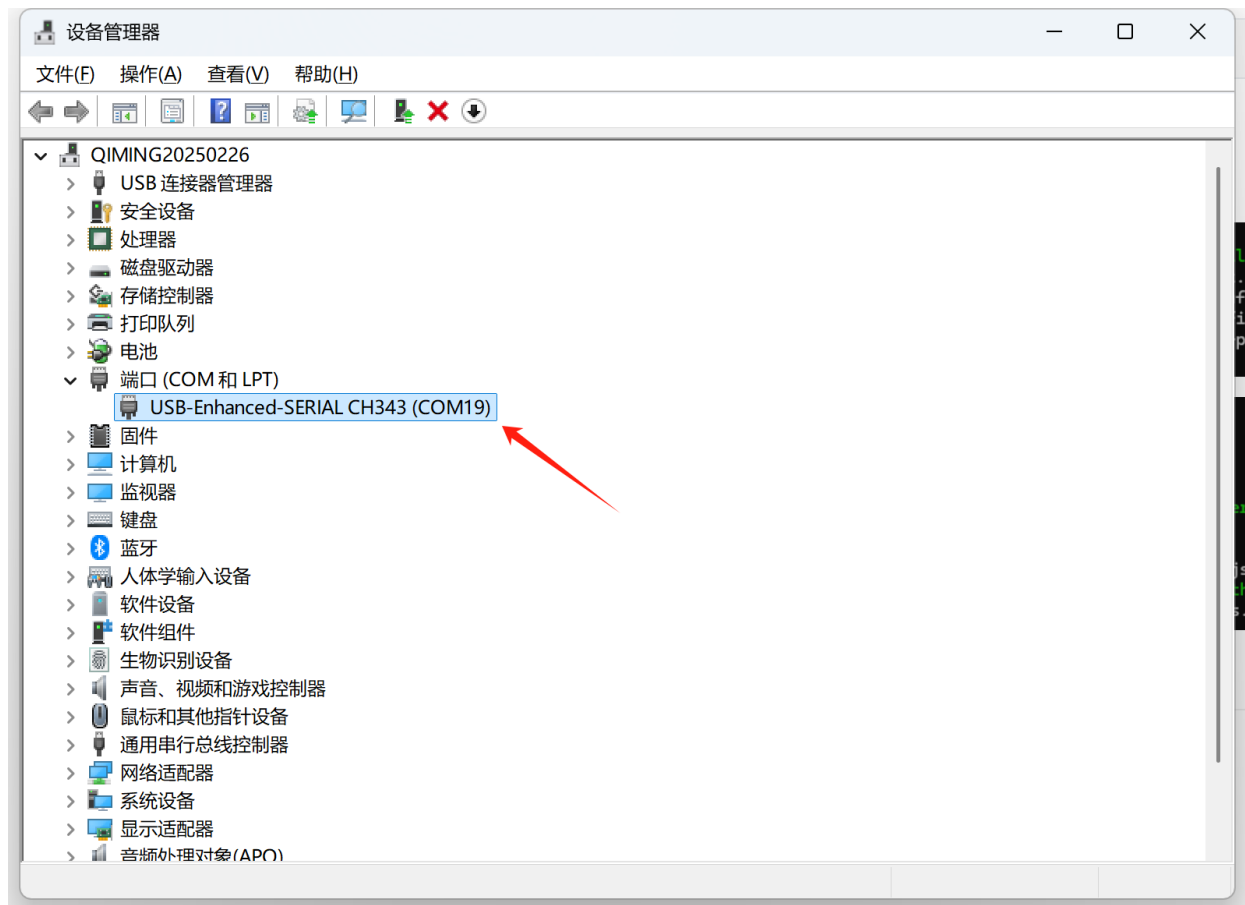
```
qiming@QIMING20250226:~/ESP_5_4/esp-idf$ ls
CMakeLists.txt  LICENSE  SECURITY.md  conftest.py  export.ps1  install.sh
COMPATIBILITY.md  README.md  SUPPORT_POLICY.md  docs  export.sh  pytest.ini
COMPATIBILITY_CN.md  README_CN.md  SUPPORT_POLICY_CN.md  examples  install.bat  sdkconfig.rename
CONTRIBUTING.md  ROADMAP.md  add_path.sh  export.bat  install.fish  sgconfig.yml
Kconfig  ROADMAP_CN.md  components  export.fish  install.ps1  sonar-project.properties
qiming@QIMING20250226:~/ESP_5_4/esp-idf$

qiming@QIMING20250226:~/ESP_5_4/esp-idf/tools$ ls
__pycache__  detect_python.sh  idf.py  mass_mfg  templates
activate.py  docker  idf_monitor.py  mkdfu.py  test_apps
ble  eclipse-code-style.xml  idf_py_actions  mkuf2.py  test_build_system
bt  esp_app_trace  idf_size.py  mocks  test_idf_py
catch  esp_prov  idf_tools.py  python_version_checker.py  test_idf_size
check_python_dependencies.py  export_utils  idfx  requirements  test_idf_tools
check_term.py  format.sh  install_util.py  requirements.json  test_mkdfu
ci  gdb_panic_server.py  kconfig_new  requirements_schema.json  tools.json
cmake  gen_esp_err_to_name.py  ldgen  set-submodules-to-github.sh  tools_schema.json
detect_python.fish  gen_soc_caps_kconfig  legacy_exports  split_paths_by_spaces.py  unit-test-app
```

5.3.2 idfx 使用方法

可参考链接: <https://github.com/abobija/idxf>

首先在设备管理器找到端口号:



执行命令:

```
idxf flash com19
```

即可正常烧录;

当然可以使用:

```
idxf all com4
```

这个命令可以完成编译，烧录，日志打印一键完成；

如果只是想要查看日志，可以执行命令：

```
idfx monitor com4
```

关于idfx的命令使用方法，可以执行：

```
idfx help
```

查看使用方法，也可以在github仓库查看使用例子。

以下是参考链接：

<https://github.com/abobija/idfx>

>>自此，开发环境基本搭建完毕。